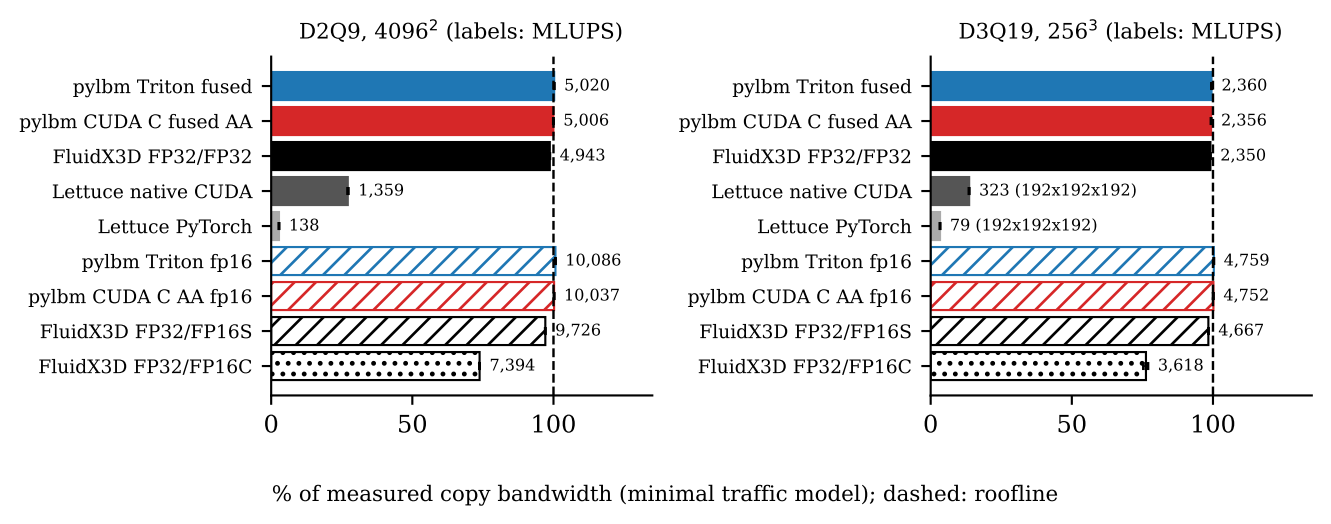
D2Q9, 4096² (labels: MLUPS)
pylbm Triton fused
pylbm CUDA C fused AA
FluidX3D FP32/FP32
Lettuce native CUDA
Lettuce PyTorch
pylbm Triton fp16
pylbm CUDA C AA fp16
FluidX3D FP32/FP16S
FluidX3D FP32/FP16C
5,020
5,006
4,943
1,359
138
10,086
10,037
9,726
7,394
0
50
100
D3Q19, 256³ (labels: MLUPS)
pylbm Triton fused
pylbm CUDA C fused AA
FluidX3D FP32/FP32
Lettuce native CUDA
Lettuce PyTorch
pylbm Triton fp16
pylbm CUDA C AA fp16
FluidX3D FP32/FP16S
FluidX3D FP32/FP16C
2,360
2,356
2,350
323 (192x192x192)
79 (192x192x192)
4,759
4,752
4,667
3,618
0
50
100
% of measured copy bandwidth (minimal traffic model); dashed: roofline